

**EE431 Digital Communications**  
**MATLAB Project Assignment 1**

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## Contents

|          |                                                                                         |           |
|----------|-----------------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                                     | <b>1</b>  |
| <b>2</b> | <b>Part 1</b>                                                                           | <b>1</b>  |
| 2.1      | Part (a): Finding $k$ and Plot of the Joint PDF . . . . .                               | 1         |
| 2.2      | Part (b): Analytical Derivation of $f_Z(z)$ and Plot . . . . .                          | 2         |
| 2.2.1    | Piecewise Form . . . . .                                                                | 2         |
| 2.3      | Part (c): Simulation, Histogram Comparison, and Comments . . . . .                      | 4         |
| 2.3.1    | Discussion and Observations . . . . .                                                   | 5         |
| 2.4      | Part (d): Signal Power, Quantization Distortion, and SQNR . . . . .                     | 5         |
| 2.4.1    | Signal Power $E[Z^2]$ . . . . .                                                         | 5         |
| 2.4.2    | Uniform Quantizer with $N = 8$ Levels on $[-1, 1]$ . . . . .                            | 5         |
| 2.4.3    | SQNR . . . . .                                                                          | 6         |
| 2.5      | Part (e): Uniform Quantization on 100000 Samples ( $N = 8$ ) and Comparison             | 6         |
| 2.6      | Part (f): Empirical PMF of Uniform Quantizer Outputs ( $N = 8$ ) . . . . .              | 6         |
| 2.7      | Part (g): Lloyd–Max Quantization on 100000 Samples ( $N = 8$ ) . . . . .                | 7         |
| 2.8      | Part (h): Empirical PMF of Lloyd–Max Quantizer Outputs ( $N = 8$ ) . . .                | 7         |
| 2.9      | Part (i): Repeating (d)–(h) for $N = 32$ and Discussion . . . . .                       | 8         |
| 2.9.1    | Analytical Approximation for Uniform Quantization ( $N = 32$ ) . .                      | 8         |
| 2.9.2    | Empirical Results and PMFs ( $N = 32$ ) . . . . .                                       | 8         |
| 2.9.3    | Discussion . . . . .                                                                    | 9         |
| <b>3</b> | <b>Part II: Lempel–Ziv–Welch (LZW) Coding</b>                                           | <b>10</b> |
| 3.1      | Part (a): Text Selection and Preprocessing . . . . .                                    | 10        |
| 3.2      | Part (b): LZW Encoding with the Given Initial Dictionary and Binary<br>Output . . . . . | 10        |
| 3.3      | Part (c): Average Codeword Length and Comparison with No Coding . .                     | 10        |
| 3.4      | Part (d): LZW Decoding and Verification . . . . .                                       | 11        |
| 3.5      | Part (e): Character PMF, Entropy, and Comparison . . . . .                              | 12        |
| <b>4</b> | <b>References</b>                                                                       | <b>13</b> |

## 1 Introduction

### 2 Part 1

#### 2.1 Part (a): Finding $k$ and Plot of the Joint PDF

The support region is the triangular set

$$\mathcal{R} = \{(x, y) : x > 0, y > 0, x + y \leq 1\},$$

with vertices  $(0, 0)$ ,  $(1, 0)$ , and  $(0, 1)$ . Since  $f_{X,Y}(x, y) = k$  over  $\mathcal{R}$ , normalization gives

$$1 = \iint_{\mathcal{R}} f_{X,Y}(x, y) dx dy = \iint_{\mathcal{R}} k dx dy = k \cdot \text{Area}(\mathcal{R}).$$

Because  $\text{Area}(\mathcal{R}) = \frac{1}{2}$ ,

$$k \cdot \frac{1}{2} = 1 \implies k = 2.$$

Hence, the joint PDF is constant with value 2 over  $\mathcal{R}$  and 0 elsewhere.

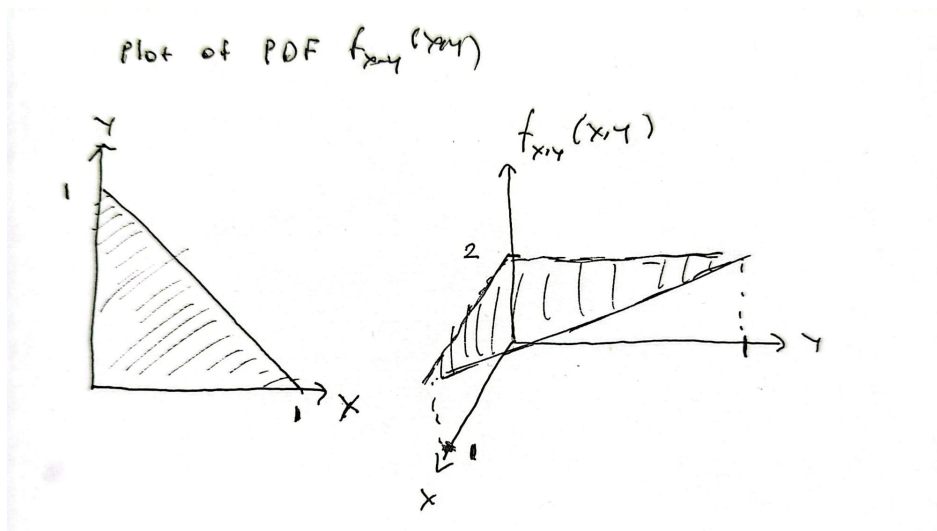


Figure 1: Joint PDF  $f_{X,Y}(x, y)$ : constant value 2 on  $x > 0, y > 0, x + y \leq 1$ , and 0 elsewhere. MATLAB

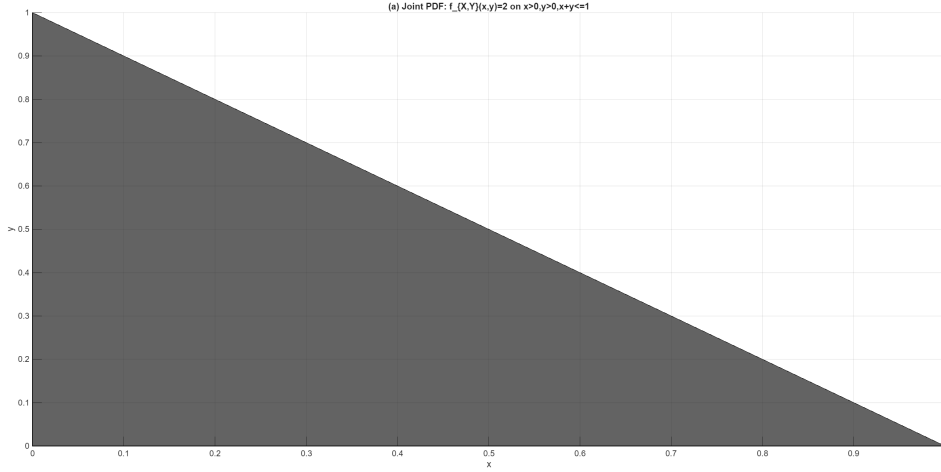


Figure 2: Joint PDF  $f_{X,Y}(x,y)$ : constant value 2 on  $x > 0$ ,  $y > 0$ ,  $x + y \leq 1$ , and 0 elsewhere.

## 2.2 Part (b): Analytical Derivation of $f_Z(z)$ and Plot

Let  $Z = X - Y$ . Use the transformation  $(x, y) \mapsto (z, y)$  defined by

$$z = x - y, \quad x = z + y.$$

The Jacobian determinant is

$$\left| \frac{\partial(x, y)}{\partial(z, y)} \right| = \begin{vmatrix} \frac{\partial x}{\partial z} & \frac{\partial x}{\partial y} \\ \frac{\partial y}{\partial z} & \frac{\partial y}{\partial y} \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1.$$

Therefore,

$$f_{Z,Y}(z, y) = f_{X,Y}(z + y, y).$$

For a fixed  $z$ , the constraints for  $(x, y) \in \mathcal{R}$  become:

$$\begin{aligned} y &> 0, \\ x = z + y > 0 &\implies y > -z, \\ x + y = z + 2y \leq 1 &\implies y \leq \frac{1 - z}{2}. \end{aligned}$$

Hence,

$$y \in \left[ \max(0, -z), \frac{1 - z}{2} \right].$$

Using  $f_{X,Y}(x, y) = 2$  on the support,

$$f_Z(z) = \int f_{Z,Y}(z, y) dy = \int_{\max(0, -z)}^{(1-z)/2} 2 dy = 2 \left( \frac{1 - z}{2} - \max(0, -z) \right).$$

### 2.2.1 Piecewise Form

$$-1 \leq z \leq 0$$

Here  $-z \geq 0$  so  $\max(0, -z) = -z$ :

$$f_Z(z) = 2 \left( \frac{1 - z}{2} + z \right) = 1 + z.$$

$0 \leq z \leq 1$  Here  $\max(0, -z) = 0$ :

$$f_Z(z) = 2 \left( \frac{1-z}{2} \right) = 1 - z.$$

Thus,

$$f_Z(z) = \begin{cases} 1 + z, & -1 \leq z \leq 0, \\ 1 - z, & 0 \leq z \leq 1, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Equivalently,

$$f_Z(z) = 1 - |z|, \quad |z| \leq 1. \quad (2)$$

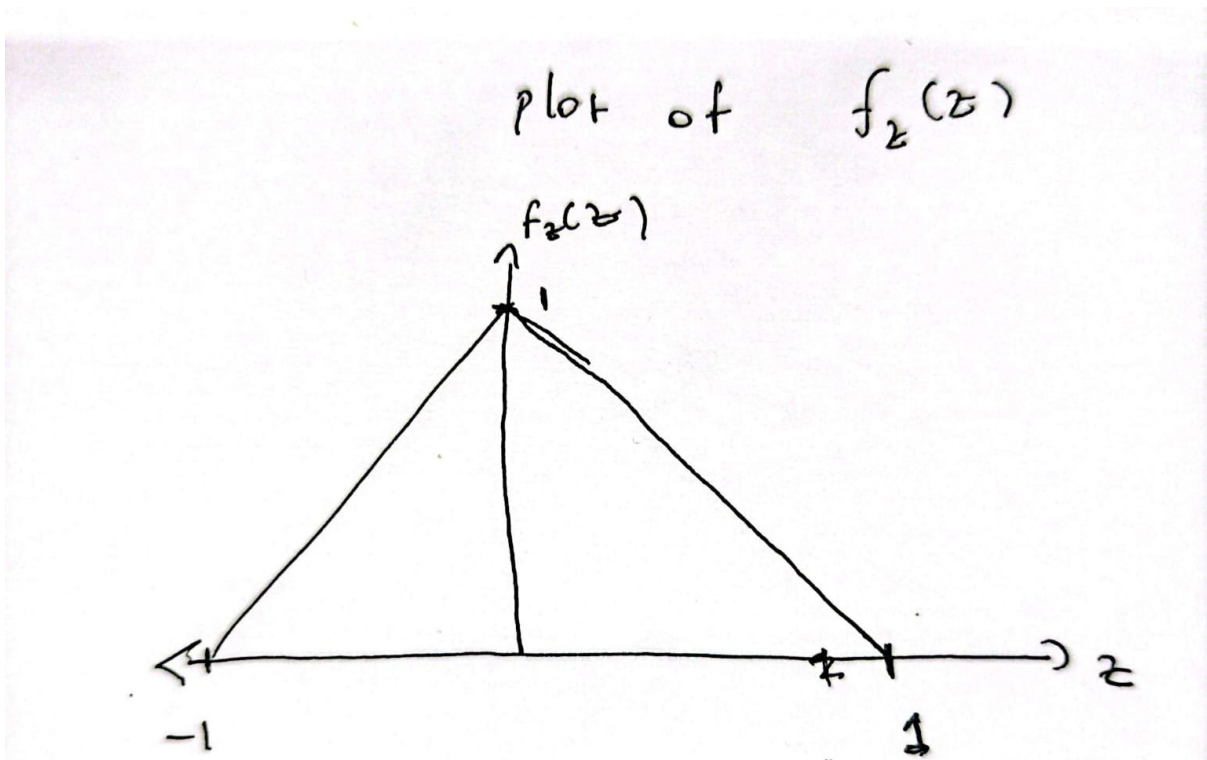


Figure 3: PDF  $f_Z(z) = 1 - |z|$  for  $|z| \leq 1$ .

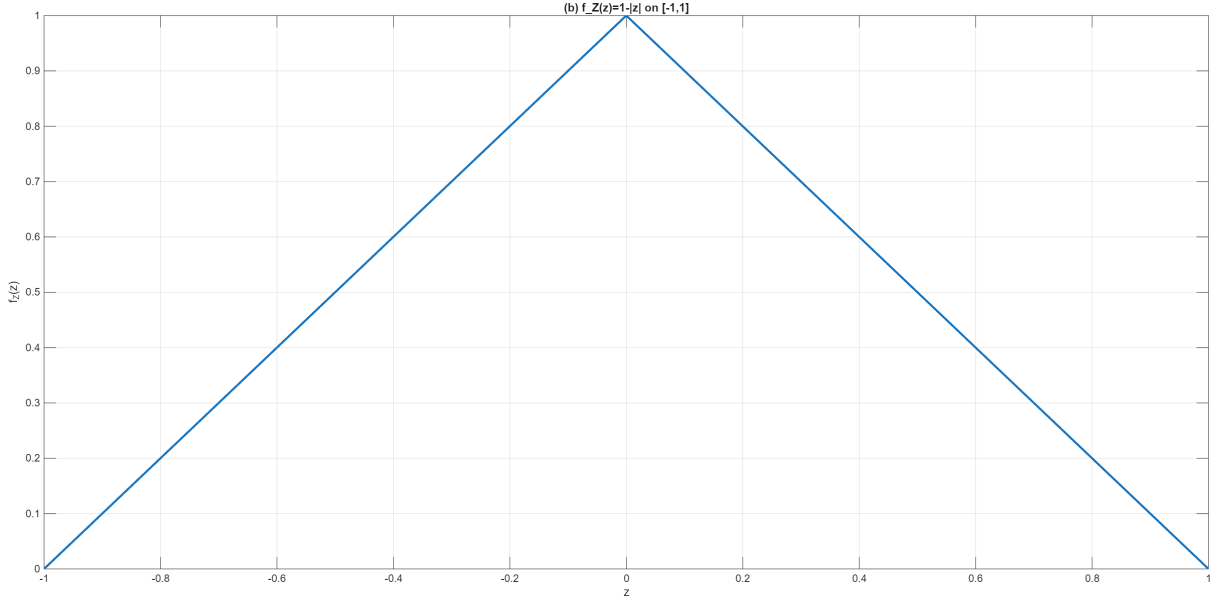


Figure 4: MATLAB PDF  $f_Z(z) = 1 - |z|$  for  $|z| \leq 1$ .

### 2.3 Part (c): Simulation, Histogram Comparison, and Comments

To verify what I have found in the analytical results,  $(X, Y)$  pairs are generated uniformly over  $\mathcal{R}$ , and then  $Z = X - Y$  is computed in MATLAB environment.

Two experiments are performed as asked and those are:

- $N = 100$  realizations of  $Z$ .
- $N = 100000$  realizations of  $Z$ .

For each case, a histogram of  $Z$  with 32 bins is plotted and compared to the theoretical curve that is calculated in part b in (2).

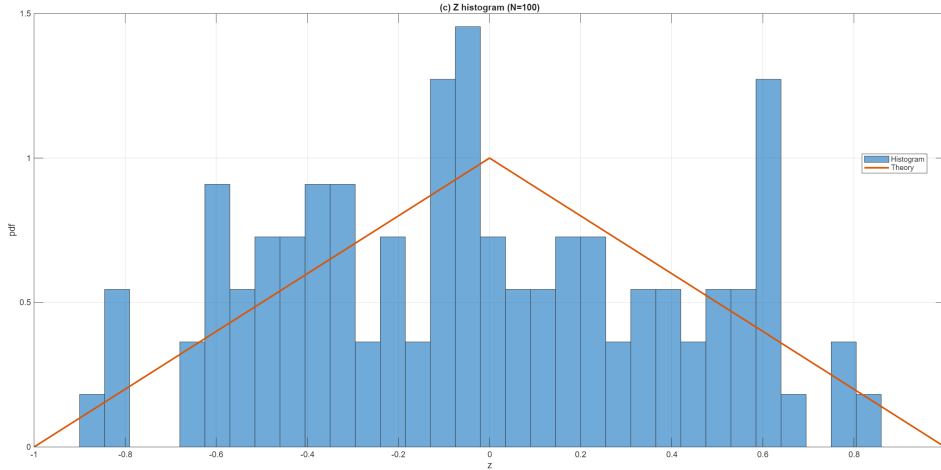


Figure 5: Histogram of  $Z$  (32 bins) for  $N = 100$ , overlaid with theoretical  $f_Z(z) = 1 - |z|$ .

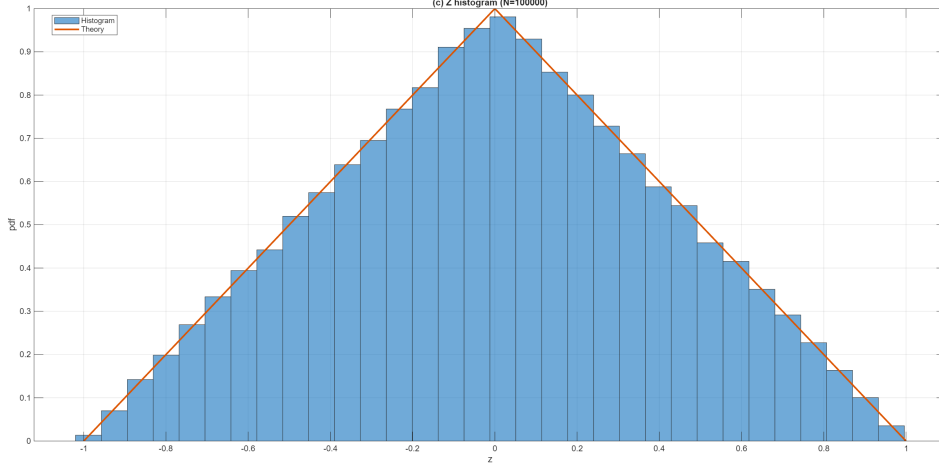


Figure 6: Histogram of  $Z$  (32 bins) for  $N = 100000$ , overlaid with theoretical  $f_Z(z) = 1 - |z|$ .

### 2.3.1 Discussion and Observations

For  $N = 100$ , the histogram is noticeably noisy and deviates from the triangular theoretical curve due to low realizations of samples. For  $N = 100000$ , the histogram matches  $f_Z(z)$  across  $[-1, 1]$ , which shows the convergence of the empirical distribution to the theoretical distribution as sample size increases.

## 2.4 Part (d): Signal Power, Quantization Distortion, and SQNR

### 2.4.1 Signal Power $E[Z^2]$

Using  $f_Z(z) = 1 - |z|$  on  $[-1, 1]$  and symmetry,

$$\begin{aligned} E[Z^2] &= \int_{-1}^1 z^2(1 - |z|) dz = 2 \int_0^1 z^2(1 - z) dz \\ &= 2 \left( \int_0^1 z^2 dz - \int_0^1 z^3 dz \right) = 2 \left( \frac{1}{3} - \frac{1}{4} \right) = \frac{1}{6}. \end{aligned}$$

Therefore,

$$E[Z^2] = \frac{1}{6} \approx 0.1667.$$

### 2.4.2 Uniform Quantizer with $N = 8$ Levels on $[-1, 1]$

A uniform quantizer with  $N = 8$  levels over  $[-1, 1]$  has step size

$$\Delta = \frac{2}{N} = \frac{2}{8} = 0.25.$$

Since  $Z$  is supported on  $[-1, 1]$ , there is no overload distortion. Using the standard approximation for uniform quantization MSE,

$$D = E[(Z - Q(Z))^2] \approx \frac{\Delta^2}{12} = \frac{0.25^2}{12} = \frac{1}{192} \approx 0.005208.$$

Thus,

$$D \approx \frac{1}{192} \approx 0.005208.$$

### 2.4.3 SQNR

The approximate signal-to-quantization-noise ratio is

$$\text{SQNR} \approx \frac{E[Z^2]}{D} = \frac{1/6}{1/192} = 32,$$

and in decibels,

$$\text{SQNR}_{\text{dB}} = 10 \log_{10}(32) \approx 15.05 \text{ dB}.$$

### 2.5 Part (e): Uniform Quantization on 100000 Samples ( $N = 8$ ) and Comparison

Using the 100000 realizations of  $Z$  we generated in part c, a uniform quantizer with  $N = 8$  levels on the interval  $[-1, 1]$  is applied. Let the quantizer output be  $Q(Z)$ . The following sample quantities are computed:

$$\hat{P}_Z = \frac{1}{M} \sum_{m=1}^M Z_m^2, \quad \hat{D} = \frac{1}{M} \sum_{m=1}^M (Z_m - Q(Z_m))^2,$$

$$\widehat{\text{SQNR}} = \frac{\hat{P}_Z}{\hat{D}}, \quad \widehat{\text{SQNR}}_{\text{dB}} = 10 \log_{10}(\widehat{\text{SQNR}}),$$

where  $M = 100000$ .

For uniform quantization over  $[-1, 1]$  with  $N$  levels, the step size is  $\Delta = \frac{2}{N}$ . For  $N = 8$ ,  $\Delta = 0.25$ . The approximate analytical MSE distortion from Part (d) is

$$D \approx \frac{\Delta^2}{12} = \frac{0.25^2}{12} = \frac{1}{192} \approx 0.0052083,$$

and using  $E[Z^2] = \frac{1}{6}$  gives

$$\text{SQNR} \approx \frac{1/6}{1/192} = 32 \quad (\approx 15.05 \text{ dB}).$$

The MATLAB results closely match these analytical values due to the large sample size.

```
===== N = 8 =====
(d) Theory approx: Delta=0.250000, D=0.005208333333, SQNR=32.000000 (15.0515 dB)
(e) Uniform sample: P_hat=0.1667926264, D_hat=0.005209103997, SQNR=32.019446 (15.0541 dB)
```

Figure 7: Console output for calculations for part d and e.

### 2.6 Part (f): Empirical PMF of Uniform Quantizer Outputs ( $N = 8$ )

The resulting PMF is plotted in Fig. 8. The PMF is symmetric around zero and has the highest probabilities for levels near zero. This is similar to triangular pdf for  $f_Z(z) = 1 - |z|$  which places more probability mass near  $z = 0$  than near the edges  $z = \pm 1$ .

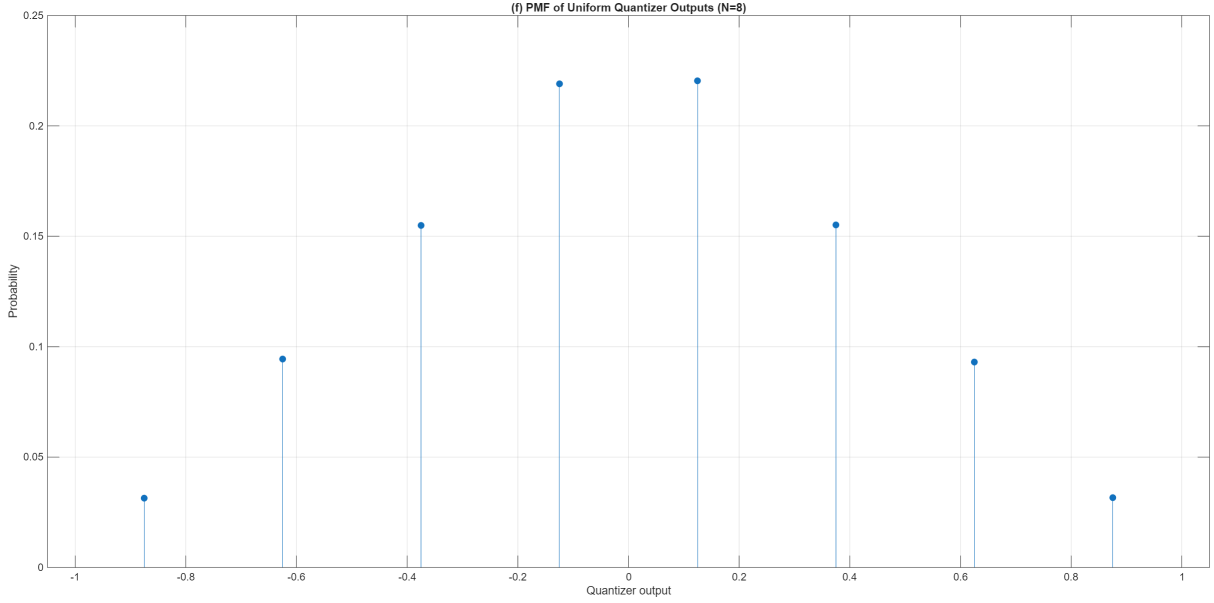


Figure 8: (f) PMF of quantizer outputs for the uniform quantizer with  $N = 8$  on  $[-1, 1]$ .

## 2.7 Part (g): Lloyd–Max Quantization on 100000 Samples ( $N = 8$ )

Using the same 100000 realizations of  $Z$ , a Lloyd–Max quantizer with  $N = 8$  levels is designed and applied. Lloyd–Max quantization iteratively updates:

- Decision boundaries as midpoints of adjacent reconstruction levels.
- Reconstruction levels as centroids of samples within each cell,

After convergence, the empirical MSE distortion and SQNR are computed via

$$\hat{D}_{\text{LM}} = \frac{1}{M} \sum_{m=1}^M (Z_m - Q_{\text{LM}}(Z_m))^2, \quad \widehat{\text{SQNR}}_{\text{LM}} = \frac{\hat{P}_Z}{\hat{D}_{\text{LM}}}.$$

Compared to the uniform quantizer, the Lloyd–Max quantizer achieves a smaller MSE and therefore a larger SQNR, since it allocates finer resolution in regions where  $Z$  is more probable.

```
===== N = 8 =====
(d) Theory approx: Delta=0.250000, D=0.005208333333, SQNR=32.000000 (15.0515 dB)
(e) Uniform sample: P_hat=0.1667926264, D_hat=0.005209103997, SQNR=32.019446 (15.0541 dB)
(g) Lloyd-Max: iters=56, D_hat=0.004087867951, SQNR=40.801863 (16.1068 dB)
```

Figure 9: (g) Lloyd–Max quantization performance for  $N = 8$  using 100000 samples: MSE distortion and SQNR compared against uniform quantization and analytical solution.

## 2.8 Part (h): Empirical PMF of Lloyd–Max Quantizer Outputs ( $N = 8$ )

The PMF is shown in Fig. 10. It remains symmetric around zero, but differs from the uniform-quantizer PMF because Lloyd–Max uses non-uniform cell widths more, narrower cells near 0, changing how frequently each reconstruction level occurs. The comparison with Part (f) illustrates that Lloyd–Max redistributes representation levels to better match the source distribution.

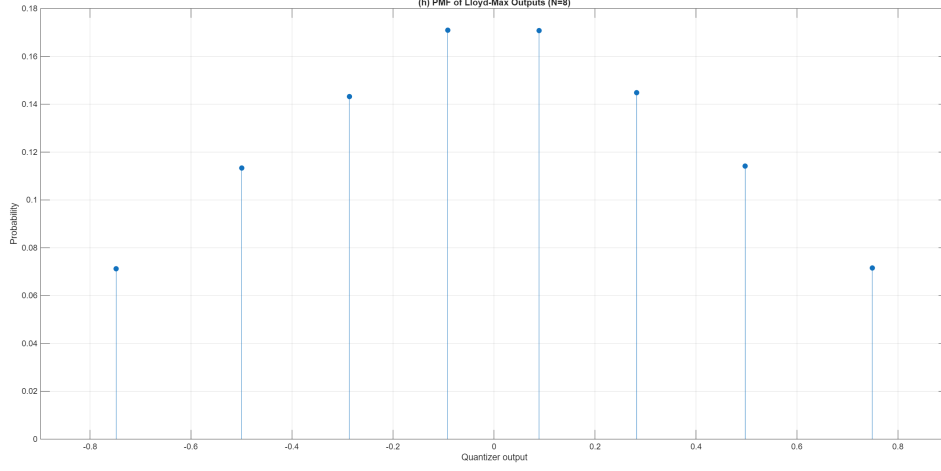


Figure 10: (h) Empirical PMF of quantizer outputs for the Lloyd–Max quantizer with  $N = 8$  on  $[-1, 1]$ .

## 2.9 Part (i): Repeating (d)–(h) for $N = 32$ and Discussion

Parts (d)–(h) are repeated for  $N = 32$ .

### 2.9.1 Analytical Approximation for Uniform Quantization ( $N = 32$ )

For  $N = 32$  levels over  $[-1, 1]$ , the step size is  $\Delta = \frac{2}{32} = 0.0625$ . The approximate uniform-quantization distortion is

$$D \approx \frac{\Delta^2}{12} = \frac{0.0625^2}{12} = \frac{1}{3072} \approx 3.2552 \times 10^{-4}.$$

Using  $E[Z^2] = \frac{1}{6}$  gives

$$\text{SQNR} \approx \frac{1/6}{1/3072} = 512 \quad (\approx 27.09 \text{ dB}).$$

Relative to  $N = 8$ , the ideal uniform-quantization SQNR improvement is

$$10 \log_{10} \left( \frac{512}{32} \right) = 10 \log_{10}(16) \approx 12.04 \text{ dB}.$$

### 2.9.2 Empirical Results and PMFs ( $N = 32$ )

Using the same 100000 samples, the empirical power, MSE, and SQNR are computed for both the uniform and Lloyd–Max quantizers (as in Parts (e) and (g)). The empirical results remain close to the analytical approximation for the uniform quantizer, and Lloyd–Max again yields a lower MSE and higher SQNR than uniform quantization.

```
===== N = 32 =====
(d) Theory approx: Delta=0.062500, D=0.000325520833, SQNR=512.000000 (27.0927 dB)
(e) Uniform sample: P_hat=0.1667926264, D_hat=0.000326218363, SQNR=511.291348 (27.0867 dB)
(g) Lloyd-Max: iters=291, D_hat=0.000269889103, SQNR=618.004300 (27.9099 dB)
```

Figure 11: Console outputs for parts d, e and f for  $N=32$  case.

The empirical PMFs for uniform and Lloyd–Max outputs with  $N = 32$  are shown in Figs. 12 and 13. Compared to  $N = 8$ , both PMFs appear smoother with more output levels, and the quantization error is significantly smaller.

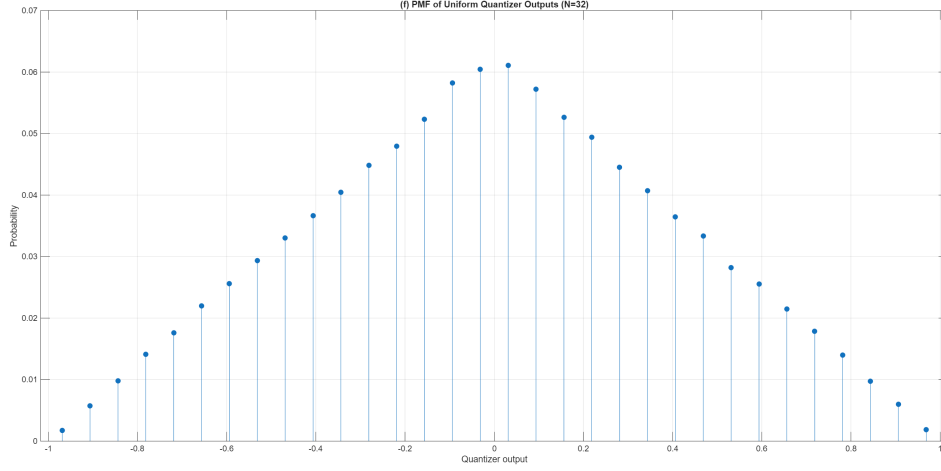


Figure 12: (i-f) Empirical PMF of uniform quantizer outputs for  $N = 32$ .

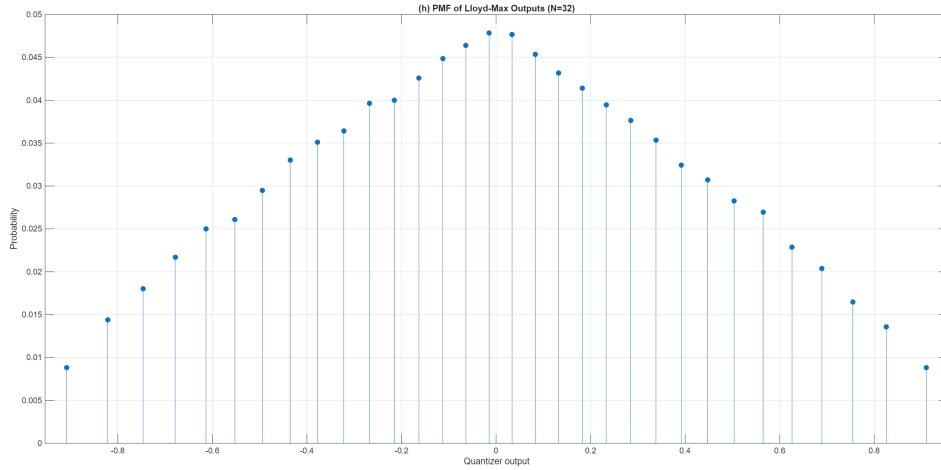


Figure 13: (i-h) Empirical PMF of Lloyd–Max quantizer outputs for  $N = 32$ .

### 2.9.3 Discussion

Increasing the number of quantization levels from  $N = 8$  to  $N = 32$  reduces the quantization step size and significantly decreases MSE distortion, leading to a large SQNR improvement which is about 12 dB. Lloyd–Max quantization provides additional improvement over uniform quantization by adapting the decision regions and representation levels to the empirical distribution of  $Z$ , although the performance gap between Lloyd–Max and uniform quantization becomes smaller as  $N$  increases.

### 3 Part II: Lempel–Ziv–Welch (LZW) Coding

#### 3.1 Part (a): Text Selection and Preprocessing

A lorem ipsum text generator[1] used to generate a 1000 word dummy text. Preprocessing for the text has been applied and resulting alphabet for our LZW has become.

$$\{\#, A, B, \dots, Z, \text{SPACE}\},$$

where  $\#$  is as an end-of-message symbol as said. After preprocessing, the symbol  $\#$  is appended to the end of the sequence to mark message termination.

```
--- Part II ---  
(a) Processed text length + "#": 6474 characters
```

Figure 14: Number of characters that is is the text. Matlab console output

#### 3.2 Part (b): LZW Encoding with the Given Initial Dictionary and Binary Output

The initial dictionary is constructed as:

$$\# \rightarrow 0, \quad A \rightarrow 1, \quad B \rightarrow 2, \quad \dots, \quad Z \rightarrow 26, \quad \text{SPACE} \rightarrow 27.$$

LZW encoding is performed by parsing the input into phrases. Let  $w$  denote the current phrase and  $c$  the next input character. At each step:

- If the concatenation  $wc$  exists in the dictionary, update  $w \leftarrow wc$ .
- Otherwise, output the dictionary index of  $w$ , add  $wc$  to the dictionary as a new entry, and set  $w \leftarrow c$ .

After the full input is processed, the index of the final phrase is output.

The sequence of output indices is then converted into a binary sequence. If the current dictionary size at the moment of output is  $D$ , the number of bits used for that output index is chosen as

$$b = \lceil \log_2(D) \rceil.$$

Therefore, the encoder produces a variable length binary bitstream whose codeword length adapts to the growing dictionary size.

```
(b) Encoded 2243 codewords, total bits = 23041
```

Figure 15: Console output that is indicating number of encoded words and total bits.

#### 3.3 Part (c): Average Codeword Length and Comparison with No Coding

Let the encoder output  $K$  integer codewords, and let the corresponding bit-lengths be  $b_1, b_2, \dots, b_K$  where  $b_k = \lceil \log_2(D_k) \rceil$  at the time of the  $k$ -th output. The total number of compressed bits is

$$B_{\text{LZW}} = \sum_{k=1}^K b_k.$$

The average number of bits per LZW codeword is

$$L_{\text{cw}} = \frac{B_{\text{LZW}}}{K}.$$

Since the input is a character sequence, it is also meaningful to report the effective number of bits per character:

$$L_{\text{char}} = \frac{B_{\text{LZW}}}{M}.$$

For comparison, without any coding, the required number of bits per character is

$$L_0 = \lceil \log_2(28) \rceil = 5 \text{ bits/character.}$$

Compression is achieved if  $L_{\text{char}} < 5$ . A way to measure is the compression ratio relative to the fixed 5-bit representation:

$$\text{CR} = \frac{5}{L_{\text{char}}}.$$

```
(c) Avg bits/codeword (LZW) = 10.2724
    Effective bits/char (LZW) = 3.5590
    Without coding: bits/char = 5
    Compression ratio vs fixed 5-bit: 1.4049
```

Figure 16: MATLAB Console Output. Calculating bits/char

### 3.4 Part (d): LZW Decoding and Verification

The LZW decoder starts from the same initial dictionary:

$$\# \rightarrow 0, A \rightarrow 1, \dots, Z \rightarrow 26, \text{SPACE} \rightarrow 27.$$

It reads the compressed bitstream sequentially. At each step, it uses the same bit-length rule as the encoder:

$$b = \lceil \log_2(D) \rceil,$$

where  $D$  is the current dictionary size used for parsing the next codeword. After converting the next  $b$  bits to an integer index, the decoder outputs the corresponding dictionary string and updates the dictionary in the position in which the received code equals the next dictionary index, the decoder reconstructs it using the previously known previous output phrase.

The decoded character sequence is compared with the processed text from Part (a), and perfect recovery is confirmed if they are identical.

```
(d) Decoding exact match = 1 (1 is success)
```

Figure 17: Console output that checks if the decoding is success.

### 3.5 Part (e): Character PMF, Entropy, and Comparison

The probability of each possible character is estimated by frequency of occurrence in the processed text. For symbol  $i \in \{0, 1, \dots, 27\}$ , let  $n_i$  be its count in the message and  $M$  be the total length. Then

$$p_i = \frac{n_i}{M}, \quad \sum_{i=0}^{27} p_i = 1.$$

The empirical PMF is plotted as a bar chart over the 28 symbols.

Using this PMF, the entropy is computed as

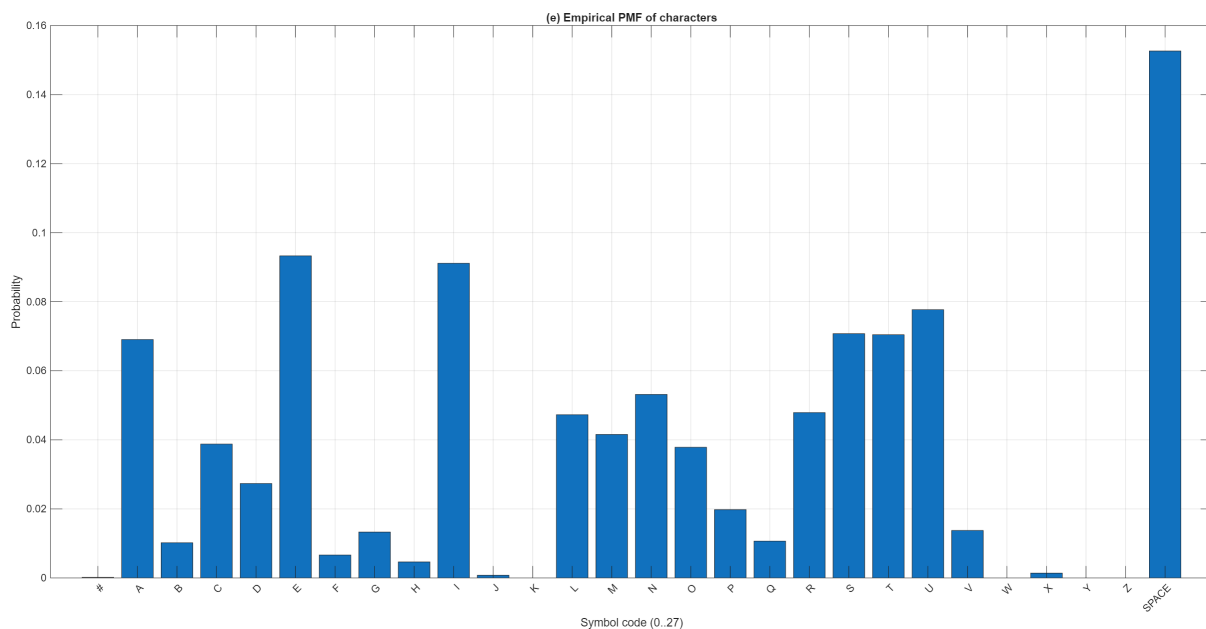
$$H = - \sum_{i=0}^{27} p_i \log_2(p_i) \quad (\text{bits/character}),$$

where terms with  $p_i = 0$  contribute 0.

Comparing the result we got from the part c we can see that LZW has compressed impressive amount of data which is about 3.56 bits/char. Which proves what we already know that compressing repeatable patten reduces the amount of data that is needed to transmit.

```
(e) Entropy H = 4.001063 bits/char
Comparing: H to LZW. Here LZW = 3.559005
```

Figure 18: Console Output. Entropy Calculations



## 4 References

[1] “Lorem Ipsum - All the facts - Ipsum generator,” Ipsum.com. [Online]. Available: <https://www.ipsum.com/> . [Accessed: Feb. 25, 2026].